Hybrid LSMC–PDE for a Generalized Gatheral Double Mean-Reverting Model

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Abstract

We study Bermudan put pricing under a generalized Gatheral double mean-reverting stochastic-volatility model. The model replaces the square-root volatility-of-volatility terms by Constant Elasticity of Variance (CEV)-type powers while preserving the double mean-reverting variance structure. The CEV terms create a boundary issue at zero: for exponents below one, the variance diffusion coefficients are not Lipschitz at the boundary, while the model is meaningful only if the variance factors remain nonnegative. We prove strong well-posedness and nonnegativity for elasticity exponents in $[\mathbf{1/2}, \mathbf{1}]$. We then derive the conditional representation needed to use the Hybrid Least-Squares Monte Carlo–Partial Differential Equation (LSMC–PDE) method in this model. The variance equations do not involve the asset price, so over one Bermudan step we condition on the Brownian increments driving the variance factors. After projecting the asset Brownian motion onto these drivers, at most one residual Brownian component remains. This gives a one-dimensional conditional equation in log-price space and an explicit Gaussian representation. These formulas provide the model-specific input for the hybrid regression convergence theorem and for the Fast Fourier Transform (FFT)-based implementation. Numerical experiments for Bermudan puts use a fixed parameter set based on a calibrated double mean-reverting specification from the literature. The direct hybrid estimator is compared with plain Least-Squares Monte Carlo (LSMC) by varying both the Euler step count and the Monte Carlo path budget.

Keywords: Bermudan option pricing, stochastic volatility, Hybrid LSMC–PDE, least-squares Monte Carlo, generalized Gatheral model

1 Introduction

Bermudan option pricing under stochastic volatility requires repeated approximation of conditional continuation values. Deterministic Partial Differential Equation (PDE) methods handle this recursion well in low-dimensional Markovian models, but their cost grows quickly when several volatility factors are present. Simulation avoids a full state-space grid, but leaves a regression problem. In stochastic-volatility models that regression depends on both the asset level and the volatility state.

We study Bermudan put pricing under a generalized Gatheral-type double mean-reverting stochastic-volatility model. The instantaneous variance v mean-reverts to a stochastic variance level v', while v' mean-reverts to a long-run level. We replace the square-root volatility-of-volatility terms by Constant Elasticity of Variance (CEV)-type powers v^{δ_1} and $(v')^{\delta_2}$, with $\delta_1, \delta_2 \in [1/2, 1]$. The square-root model is recovered when $\delta_1 = \delta_2 = 1/2$.

The CEV generalization creates a boundary issue at zero. For exponents below one, the variance diffusion coefficients are not Lipschitz there, while the model is meaningful only as long as the variance factors remain nonnegative. We prove strong existence, pathwise uniqueness, and nonnegativity of the two-factor variance system. The proof uses the triangular structure of the model: first v', then v conditional on v', and finally the asset equation through its stochastic exponential.

The numerical method builds on the Hybrid Least-Squares Monte Carlo–PDE (LSMC–PDE) framework of Farahany, Jackson, and Jaimungal [1]. We use their convergence theorem as a general result and focus on the model-specific work needed for the generalized Gatheral double mean-reverting (gDMR) dynamics. Since the variance equations do not involve the asset price, we condition on the Brownian increments driving the variance factors over one Bermudan step. After projecting the asset Brownian motion onto these drivers, at most one residual Brownian component remains, and the conditional asset-price step becomes a one-dimensional equation in log-price space. This conditioning is used only inside an iterated expectation and does not enlarge the exercise filtration.

The key quantities in this reduction are the correlated Brownian shift Z_n, the integrated drift I_n, the integrated residual variance B_n, and the terminal volatility state. These path statistics are the model-specific quantities needed to use the hybrid method. We then implement the method by combining Fast Fourier Transform (FFT) propagation in the asset direction with least-squares regression over volatility states.

The work is related to regression-based Bermudan pricing methods such as Least-Squares Monte Carlo (LSMC) [2–4], mixed Monte Carlo–PDE (MC–PDE) methods for stochastic-volatility models [1, 5–7], and stochastic-volatility models with mean-reverting or CEV-type variance dynamics [8–14].

The rest of the paper is organized as follows. Section 2 introduces the model, states the Bermudan recursion, and gives the well-posedness result. Section 3 derives the conditional hybrid representation. Section 4 describes the numerical implementation, Section 5 reports the experiments, and Section 6 concludes.

2 Model, Bermudan problem, and well-posedness

This section introduces the generalized gDMR model and the Bermudan recursion. The equations for the variance factors do not involve the asset price, while the asset equation depends on the variance factors. This one-way coupling is used in Section 3 to obtain the conditional one-dimensional representation.

2.1 Generalized gDMR dynamics

Fix a finite horizon $T > 0$. Let

$$(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in[0,T]}, \mathbb{Q})$$

be a filtered probability space satisfying the usual conditions. We work under a risk-neutral measure $\mathbb{Q}$ with deterministic money-market account $B_t = e^{rt}$. The space supports a three-dimensional correlated $(\mathcal{F}_t)$-Brownian motion

$$W = (W^{(1)}, W^{(2)}, W^{(3)})$$

with constant quadratic covariations

$$\langle W^{(i)}, W^{(j)}\rangle_t = \rho_{ij} t, \qquad 1 \le i, j \le 3.$$

The matrix $\rho = (\rho_{ij})_{1\le i,j\le 3}$ is symmetric positive semidefinite and satisfies $\rho_{ii} = 1$. We assume $|\rho_{23}| < 1$, so that the covariance matrix of the two Brownian drivers of the variance factors is invertible in the projection used in Section 3.

Let $S_0 > 0$, $v_0 \ge 0$, and $v'_0 \ge 0$. We consider the generalized gDMR dynamics

$$\begin{aligned} \mathrm{d}S_t &= r\, S_t \,\mathrm{d}t + S_t\sqrt{v_t}\,\mathrm{d}W_t^{(1)}, \\ \mathrm{d}v_t &= \kappa_1\,(v'_t - v_t)\,\mathrm{d}t + \xi_1 v_t^{\delta_1}\,\mathrm{d}W_t^{(2)}, \\ \mathrm{d}v'_t &= \kappa_2\,(\theta - v'_t)\,\mathrm{d}t + \xi_2 (v'_t)^{\delta_2}\,\mathrm{d}W_t^{(3)}, \end{aligned} \tag{1}$$

where $r \ge 0$, $\kappa_1, \kappa_2, \theta, \xi_1, \xi_2 > 0$, and $\delta_1, \delta_2 \in [1/2, 1]$. The factor v is the instantaneous variance and mean-reverts to the stochastic level v', while v' mean-reverts to the long-run level θ. The exponents δ_1 and δ_2 replace the square-root volatility-of-volatility coefficients by CEV-type powers; the square-root double mean-reverting model is recovered when $\delta_1 = \delta_2 = 1/2$.

Assumption 2.1 (Pricing martingale) The discounted stock price $e^{-rt}S_t$ is a true $\mathbb{Q}$-martingale on $[0, T]$.

This martingale condition is part of the risk-neutral pricing specification. The well-posedness result below ensures that the stochastic exponential is well defined and that the variance factors remain nonnegative; it does not by itself prove the true-martingale property.

The model itself is formulated on the nonnegative variance state space

$$\mathcal{D} := (0,\infty) \times [0,\infty)^2.$$

For the proof of strong well-posedness in Subsection 2.3, we extend only the variance diffusion coefficients to the real line by setting

$$\sigma_i(x) := \xi_i (x^+)^{\delta_i}, \qquad x^+ := \max\{x, 0\}, \qquad i = 1, 2.$$

Theorem 2.4 shows that the variance factors remain nonnegative. Therefore, along the solution paths of (1), the coefficients coincide with the original coefficients on the nonnegative state space.

Remark 2.2 (One-way coupling) The equations for (v, v') do not involve S. Over one exercise interval, we may therefore simulate the Brownian increments driving the variance factors before simulating the remaining asset-price randomness. Conditioning on these increments fixes both the variance segment and the part of the asset Brownian motion correlated with the variance drivers. The only remaining asset randomness is the orthogonal Brownian component. This conditioning is used inside the one-step conditional expectation and does not change the information used for the exercise decision at the exercise dates.

2.2 Bermudan stopping problem

Let

$$\pi = \{0 = t_0 < t_1 < \cdots < t_M = T\}$$

be the Bermudan exercise grid. At t_n, the payoff is $h_n(S_{t_n})$, where $h_n : (0,\infty) \to \mathbb{R}$ is Borel. We assume

$$\mathbb{E}^{\mathbb{Q}}\left[\max_{0\le n\le M} e^{-rt_n} |h_n(S_{t_n})|\right] < \infty.$$

For the Bermudan put, $h_n(s) = (K - s)^+$.

The well-posedness result below gives a time-homogeneous Markov process on

$$\mathcal{D} = (0,\infty) \times [0,\infty)^2.$$

Thus the Bermudan value functions may be defined by backward induction. Write

$$\Delta t_n := t_{n+1} - t_n$$

and set

$$U_M(s, v, w) := h_M(s),$$

where w denotes the second variance coordinate. For $n = M-1, \dots, 0$, define

$$C_n(s, v, w) := e^{-r\Delta t_n} \mathbb{E}^{\mathbb{Q}}\left[U_{n+1}(S_{t_{n+1}}, v_{t_{n+1}}, v'_{t_{n+1}}) \,\middle|\, S_{t_n} = s,\ v_{t_n} = v,\ v'_{t_n} = w\right],$$

and

$$U_n(s,v,w) := \max\{h_n(s), C_n(s,v,w)\}.$$

The time-zero Bermudan value is

$$V_0 = U_0(S_0, v_0, v_0').$$

2.3 Well-posedness and nonnegativity

The model is written on the nonnegative variance state space. We next verify that this state space is preserved. The only coefficients that may fail to be Lipschitz are the CEV-type variance diffusions. The triangular structure of the model allows us to solve v' first, then v conditional on v', and finally the asset equation.

Assumption 2.3 (CEV exponents) Throughout this section we impose

$$\delta_1, \delta_2 \in [1/2, 1].$$

For $\delta \in [1/2, 1]$, the map $x \mapsto (x^+)^\delta$ satisfies

$$\left|(x^+)^\delta - (y^+)^\delta\right| \le |x-y|^\delta, \qquad x, y \in \mathbb{R}.$$

Hence the diffusion coefficient $x \mapsto \xi(x^+)^\delta$ satisfies the Yamada–Watanabe modulus condition [14], because the choice $\rho(u) = Cu^\delta$ gives

$$\int_0^\varepsilon \frac{\mathrm{d}u}{\rho(u)^2} = \infty \qquad \text{for } \delta \ge \frac{1}{2}.$$

The upper bound $\delta \le 1$ gives at most linear growth.

Theorem 2.4 (Strong well-posedness and nonnegativity) *Assume Assumption 2.3. For every $T > 0$ and every initial state*

$$S_0 > 0, \qquad v_0 \ge 0, \qquad v_0' \ge 0,$$

the nonnegative-state system (1) admits a unique strong solution $(S_t, v_t, v_t')_{t\in[0,T]}$ with continuous paths such that

$$S_t > 0, \qquad v_t \ge 0, \qquad v_t' \ge 0, \qquad 0 \le t \le T, \quad \mathbb{Q}\text{-a.s.}$$

Moreover, for every deterministic initial state in $\mathcal{D}$, the solution is a time-homogeneous strong Markov process on $\mathcal{D}$.

The proof is given in Appendix A. It uses the triangular structure of the variance equations: solve the v'-equation first, then the v-equation with v' as an adapted input, and finally recover the asset from its stochastic exponential. The positive-part extension and the one-dimensional invariance criterion keep both variance factors nonnegative.

The lower bound $\delta_i \ge 1/2$ is the Yamada–Watanabe threshold for pathwise uniqueness of power diffusion coefficients. We do not impose a Feller-type condition, since the theorem proves nonnegativity rather than strict positivity or boundary inaccessibility.

3 Conditional hybrid representation for the generalized gDMR model

This section derives the conditional representation used by the hybrid method. The general convergence theorem is taken from Farahany, Jackson, and Jaimungal [1]; our role here is to identify the model-specific quantities needed to apply it. The variance equations do not involve S, so over one Bermudan interval we condition on the Brownian increments driving the variance factors. After projecting the asset Brownian driver onto these increments, at most one residual Brownian component remains.

3.1 Brownian projection

Fix an interval $[t_n, t_{n+1}]$ in the Bermudan exercise grid and define

$$\mathcal{H}_n := \mathcal{F}_{t_n} \vee \sigma\Big(W_s^{(2)} - W_{t_n}^{(2)}, W_s^{(3)} - W_{t_n}^{(3)} : s \in [t_n, t_{n+1}]\Big).$$

Since v and v' are driven only by $W^{(2)}$ and $W^{(3)}$, the path $(v_s, v'_s)_{s\in[t_n,t_{n+1}]}$ is $\mathcal{H}_n$-measurable. We project $W^{(1)}$ onto the span of $(W^{(2)}, W^{(3)})$. Set

$$\Sigma_{23} := \begin{bmatrix} 1 & \rho_{23} \\ \rho_{23} & 1 \end{bmatrix}, \qquad \beta := \Sigma_{23}^{-1} \begin{bmatrix} \rho_{12} \\ \rho_{13} \end{bmatrix} = \begin{bmatrix} \beta_2 \\ \beta_3 \end{bmatrix}.$$

Equivalently,

$$\beta_2 = \frac{\rho_{12} - \rho_{13}\rho_{23}}{1 - \rho_{23}^2}, \qquad \beta_3 = \frac{\rho_{13} - \rho_{12}\rho_{23}}{1 - \rho_{23}^2}. \tag{2}$$

The residual variance is

$$\begin{aligned} \sigma_\perp^2 &:= 1 - \begin{bmatrix} \rho_{12} & \rho_{13} \end{bmatrix} \Sigma_{23}^{-1} \begin{bmatrix} \rho_{12} \\ \rho_{13} \end{bmatrix} \\ &= \frac{1 - \rho_{12}^2 - \rho_{13}^2 - \rho_{23}^2 + 2\rho_{12}\rho_{13}\rho_{23}}{1 - \rho_{23}^2} \geq 0. \end{aligned} \tag{3}$$

When $\sigma_\perp > 0$,

$$B_t^\perp := \frac{W_t^{(1)} - \beta_2 W_t^{(2)} - \beta_3 W_t^{(3)}}{\sigma_\perp}$$

is a Brownian motion, and its increments over $[t_n, t_{n+1}]$ are independent of $\mathcal{H}_n$. Hence

$$\mathrm{d}W_t^{(1)} = \beta_2 \,\mathrm{d}W_t^{(2)} + \beta_3 \,\mathrm{d}W_t^{(3)} + \sigma_\perp \,\mathrm{d}B_t^\perp. \tag{4}$$

If $\sigma_\perp = 0$, the residual term is absent and the asset Brownian driver is fully determined by the Brownian drivers of the variance factors.

3.2 Conditional one-dimensional equation

Let $X_t := \log S_t$. Substituting (4) into the asset equation gives

$$\mathrm{d}X_t = \Big(r - \tfrac{1}{2}v_t\Big)\mathrm{d}t + \sqrt{v_t}\Big(\beta_2\,\mathrm{d}W_t^{(2)} + \beta_3\,\mathrm{d}W_t^{(3)}\Big) + \sigma_\perp\sqrt{v_t}\,\mathrm{d}B_t^\perp.$$

Define the $\mathcal{H}_n$-measurable shift

$$Z_n := \int_{t_n}^{t_{n+1}} \sqrt{v_s}\Big(\beta_2\,\mathrm{d}W_s^{(2)} + \beta_3\,\mathrm{d}W_s^{(3)}\Big).$$

For $t \in [t_n, t_{n+1}]$, define also

$$I_n(t) := \int_t^{t_{n+1}} \Big(r - \tfrac{1}{2}v_s\Big)\mathrm{d}s, \qquad B_n(t) := \sigma_\perp^2 \int_t^{t_{n+1}} v_s\,\mathrm{d}s,$$

and write $I_n := I_n(t_n)$ and $B_n := B_n(t_n)$.

Proposition 3.1 (Conditional one-step representation) *Let $G : \mathbb{R}\times[0,\infty)^2 \to \mathbb{R}$ be bounded and Borel. Conditionally on $\mathcal{H}_n$, the function*

$$u(t,y) := \mathbb{E}^{\mathbb{Q}}\Big[G\Big(Y_{t_{n+1}}^{t,y} + Z_n, v_{t_{n+1}}, v'_{t_{n+1}}\Big)\,\Big|\,\mathcal{H}_n\Big],$$

where

$$\mathrm{d}Y_s^{t,y} = \Big(r - \tfrac{1}{2}v_s\Big)\mathrm{d}s + \sigma_\perp\sqrt{v_s}\,\mathrm{d}B_s^\perp, \qquad Y_t^{t,y} = y,$$

is the bounded mild solution of

$$\partial_t u + \Big(r - \tfrac{1}{2}v_t\Big)\partial_y u + \frac{1}{2}\sigma_\perp^2 v_t \partial_{yy} u = 0, \qquad u(t_{n+1}, y) = G(y + Z_n, v_{t_{n+1}}, v'_{t_{n+1}}).$$

Equivalently, for an auxiliary $\xi \sim N(0,1)$,

$$u(t,y) = \mathbb{E}_\xi\Big[G\Big(y + Z_n + I_n(t) + \sqrt{B_n(t)}\,\xi, v_{t_{n+1}}, v'_{t_{n+1}}\Big)\Big],$$

with the deterministic interpretation when $B_n(t) = 0$.

Proof. After conditioning on $\mathcal{H}_n$, the variance path and the shift Z_n are fixed. If $\sigma_\perp > 0$, the only remaining randomness is the residual Brownian motion; if $\sigma_\perp = 0$, the conditional step is deterministic. The conditional Feynman–Kac formula gives the backward Kolmogorov equation. Since the coefficients are independent of y, the terminal log-price is Gaussian with mean $y + I_n(t)$ and variance $B_n(t)$, which gives the displayed formula.

Thus, for an asset-space next-step value function F and $G(y,v,w) := F(e^y, v, w)$, the pathwise pre-surface is

$$\bar{C}_n^F(s; \mathcal{H}_n) := e^{-r\Delta t_n} u(t_n, \log s).$$

Thus the conditional asset step depends on the sampled $(W^{(2)}, W^{(3)})$-path over $[t_n, t_{n+1}]$ only through

$$Z_n, \qquad I_n, \qquad B_n, \qquad v_{t_{n+1}}, \qquad v'_{t_{n+1}},$$

while (v_{t_n}, v'_{t_n}) is the state used in the outer regression.

3.3 Connection with the hybrid convergence theorem

The convergence theorem for the truncated Hybrid LSMC–PDE regression is proved in [1]. We use it here after checking the model-specific conditions.

First, the asset equation is multiplicative. For $0 \le t < u \le T$,

$$S_u = S_t R_{t,u},$$

where

$$R_{t,u} = \exp\left(\int_t^u \left(r - \frac{1}{2}v_s\right) \mathrm{d}s + \int_t^u \sqrt{v_s}\, \mathrm{d}W_s^{(1)}\right).$$

Thus the return factor $R_{t,u}$ depends on the Brownian and variance paths, but not on the value of S_t.

Second, Proposition 3.1 shows that, on each Bermudan interval, the sampled $(W^{(2)}, W^{(3)})$-path enters the conditional asset-price step only through the finite-dimensional statistic

$$\Theta_n := (V_n, V_{n+1}, Z_n, I_n, B_n), \qquad V_n = (v_{t_n}, v'_{t_n}).$$

Because the variance equations do not involve S, the model is one-way coupled. Hence we can apply the hybrid convergence theorem of [1, Theorem 1].

With the compact volatility truncation, bounded basis functions, and nonsingular population Gram matrices required in [1], the stabilized hybrid regression coefficients converge almost surely. For each fixed asset level, the fitted continuation values converge uniformly on the chosen compact volatility domain.

4 Numerical implementation

The numerical experiments use the following implementation. The conditional representation is derived in Section 3; here we specify the grids, variance discretization, accumulated path statistics, FFT step, regression step, and time-zero estimator. These numerical choices are separate from the convergence statement quoted in Section 3.

4.1 Grids and basis functions

Let

$$\pi = \{0 = t_0 < t_1 < \cdots < t_M = T\}$$

be the Bermudan exercise grid, and let h_n denote the exercise payoff at t_n. We work on a uniform log-price grid

$$y_i = y_{\min} + (i-1)\Delta y, \qquad \Delta y = \frac{y_{\max} - y_{\min}}{N_S - 1}, \qquad i = 1, \dots, N_S,$$

and set

$$s_i = e^{y_i}, \qquad \mathcal{S}_h := \{s_1, \dots, s_{N_S}\}.$$

The interval $[y_{\min}, y_{\max}]$ is chosen large enough to contain the relevant asset values used in the computation.

For the volatility variables, write $w = v'$. Choose a compact rectangle

$$\mathcal{D}_v = [0, v_{\max}] \times [0, w_{\max}] \subset [0, \infty)^2$$

and a vector of continuous bounded basis functions

$$\phi = (\phi_1, \dots, \phi_{d_B}), \qquad \operatorname{supp}(\phi_k) \subset \mathcal{D}_v.$$

Once chosen, the domain $\mathcal{D}_v$ and the basis functions are fixed before the training regression.

At each exercise date, we approximate the continuation value by a separable regression form

$$c_n(s, v, w) \approx a_n(s) \cdot \phi(v, w), \qquad a_n(s) \in \mathbb{R}^{d_B}.$$

Numerically, the coefficient vectors are computed only at the grid points $s_i \in \mathcal{S}_h$. Off-grid asset values are evaluated by interpolation in the log-price coordinate.

4.2 Variance simulation and path statistics

Let

$$0 = \tau_0 < \tau_1 < \cdots < \tau_L = T$$

be an Euler grid containing all exercise dates. Write

$$\Delta\tau_\ell := \tau_{\ell+1} - \tau_\ell.$$

For each path j, simulate correlated Gaussian increments

$$(\Delta W_\ell^{(2),j}, \Delta W_\ell^{(3),j})$$

with variances $\Delta\tau_\ell$ and covariance $\rho_{23}\Delta\tau_\ell$. The numerical implementation uses the following positive-part Euler discretization:

$$\begin{aligned}
\bar{w}_\ell^j &:= (w_\ell^j)^+, \qquad \bar{v}_\ell^j := (v_\ell^j)^+, \\
w_{\ell+1}^j &:= \left[w_\ell^j + \kappa_2(\theta - \bar{w}_\ell^j)\Delta\tau_\ell + \xi_2(\bar{w}_\ell^j)^{\delta_2}\Delta W_\ell^{(3),j}\right]^+, \\
v_{\ell+1}^j &:= \left[v_\ell^j + \kappa_1(\bar{w}_\ell^j - \bar{v}_\ell^j)\Delta\tau_\ell + \xi_1(\bar{v}_\ell^j)^{\delta_1}\Delta W_\ell^{(2),j}\right]^+.
\end{aligned}$$

For each exercise interval $[t_n, t_{n+1}]$, the path statistics are accumulated over the Euler substeps contained in that interval:

$$\begin{aligned}
Z_n^j &:= \sum_{\tau_\ell \in [t_n, t_{n+1})} \sqrt{\bar{v}_\ell^j}\left(\beta_2 \Delta W_\ell^{(2),j} + \beta_3 \Delta W_\ell^{(3),j}\right), \\
I_n^j &:= \sum_{\tau_\ell \in [t_n, t_{n+1})} \left(r - \frac{1}{2}\bar{v}_\ell^j\right)\Delta\tau_\ell, \\
B_n^j &:= \sigma_\perp^2 \sum_{\tau_\ell \in [t_n, t_{n+1})} \bar{v}_\ell^j \Delta\tau_\ell.
\end{aligned}$$

Here β_2, β_3 are the projection coefficients in (2), and $\sigma_\perp$ is given by (3). The left-point rule is natural for Z_n^j, since Z_n^j approximates a stochastic integral evaluated with left endpoints. The same left-point values are used for I_n^j and B_n^j for consistency with the Euler simulation, and the same increments $\Delta W_\ell^{(2),j}, \Delta W_\ell^{(3),j}$ are used in the variance updates and in Z_n^j.

4.3 Pathwise FFT propagation and regression

Fix $n \in \{0, \dots, M-1\}$. Suppose that $\widehat{V}_{n+1}^N$ has already been constructed on the asset grid and is represented through the volatility basis. For each Monte Carlo path j, the simulated Brownian increments driving the variance factors on $[t_n, t_{n+1}]$ determine the endpoint state

$$(v_{t_{n+1}}^j, w_{t_{n+1}}^j)$$

and the statistics Z_n^j, I_n^j, B_n^j. Define the terminal grid function

$$g_{n+1}^j(y_i) := \widehat{V}_{n+1}^N\left(e^{y_i}, v_{t_{n+1}}^j, w_{t_{n+1}}^j\right), \qquad i = 1, \dots, N_S.$$

By the Gaussian representation in Section 3,

$$u_n^j(y) = \mathbb{E}_\xi\left[g_{n+1}^j\left(y + Z_n^j + I_n^j + \sqrt{B_n^j}\xi\right)\right], \qquad \xi \sim N(0,1).$$

With the Fourier convention

$$\mathcal{F}f(\omega) = \int_{\mathbb{R}} e^{-\mathrm{i}\omega y} f(y)\,\mathrm{d}y,$$

we obtain

$$\mathcal{F}u_n^j(\omega) = \mathcal{F}g_{n+1}^j(\omega)\exp\left(\mathrm{i}\omega Z_n^j + \mathrm{i}\omega I_n^j - \frac{1}{2}\omega^2 B_n^j\right).$$

Define

$$\Psi_n^j(\omega) := \mathrm{i}\omega Z_n^j + \mathrm{i}\omega I_n^j - \frac{1}{2}\omega^2 B_n^j.$$

On the uniform log-price grid, the FFT implementation is

$$u_n^j(\cdot) \approx \mathrm{FFT}^{-1}\left(\mathrm{FFT}[g_{n+1}^j]\,\exp(\Psi_n^j)\right), \tag{5}$$

where the exponential is applied pointwise at the discrete Fourier frequencies. The FFT input is the unshifted terminal grid g_{n+1}^j; the correlated shift Z_n^j is included only through the multiplier. Equivalently, one may shift the terminal grid first and omit $\mathrm{i}\omega Z_n^j$, but the two conventions must not be combined. If $B_n^j = 0$, the conditional propagation reduces to the deterministic log-price shift $Z_n^j + I_n^j$.

The pathwise pre-surface on the asset grid is

$$\widehat{C}_n^j(s_i) := e^{-r\Delta t_n} u_n^j(y_i), \qquad i = 1, \dots, N_S.$$

The regression step is performed at each exercise date t_n, $n = M-1, \dots, 1$, and at each asset grid point s_i. It estimates the dependence of the pathwise pre-surface $\widehat{C}_n^j(s_i)$ on the current volatility state. Define

$$V_n^j := (v_{t_n}^j, w_{t_n}^j).$$

The sample Gram matrix is

$$A_n^N := \frac{1}{N}\sum_{j=1}^{N} \phi(V_n^j)\phi(V_n^j)^\top,$$

and the right-hand side for the grid point s_i is

$$b_n^N(s_i) := \frac{1}{N}\sum_{j=1}^{N} \phi(V_n^j)\widehat{C}_n^j(s_i).$$

For numerical stability, we use the truncated inverse

$$[A]_R^{-1} := \begin{cases} A^{-1}, & \text{if } A \text{ is invertible and } \|A^{-1}\| \le R, \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

Then set

$$a_n^N(s_i) := [A_n^N]_R^{-1} b_n^N(s_i). \tag{7}$$

The fitted continuation surface is

$$\widehat{C}_n^N(s_i, v, w) := a_n^N(s_i) \cdot \phi(v, w),$$

and the Bermudan update is

$$\widehat{V}_n^N(s_i, v, w) := \max \left\{ h_n(s_i), \widehat{C}_n^N(s_i, v, w) \right\}.$$

Starting from

$$\widehat{V}_M^N(s_i, v, w) := h_M(s_i),$$

the recursion is performed backward for $n = M - 1, \ldots, 1$. The fitted functions are defined at the asset grid points and extended to off-grid asset values by log-price interpolation.

4.4 Time-zero estimator and pseudocode

At $t_0 = 0$, the initial volatility state is fixed, so no cross-sectional regression is needed. After $\widehat{V}_1^N$ has been fitted, the time-zero continuation value is evaluated on the asset grid by averaging first-step pathwise pre-surfaces:

$$\widehat{C}_0^{N_0}(s_i) := \frac{1}{N_0} \sum_{j=1}^{N_0} \widehat{C}_0^{j,\mathrm{eval}}(s_i).$$

Here N is the number of simulated variance paths used to fit the regression surfaces in the backward recursion. The number N_0 is the number of paths used only for the time-zero average above. These can be different if the time-zero average is computed from a fresh evaluation sample. In the numerical experiments we reuse the training paths for this average, so $N_0 = N$; this is the in-sample direct estimator. The averaged grid values above define $\widehat{C}_0^{N_0}$ on the asset grid; off-grid values are evaluated by the log-price interpolation convention stated above. Thus $\widehat{C}_0^{N_0}(S_0)$ denotes the averaged continuation value at the initial asset price, and the direct estimator is

$$\widehat{V}_{d,0}(S_0) = \max \left\{ h_0(S_0), \widehat{C}_0^{N_0}(S_0) \right\}.$$

Algorithm 1 Hybrid LSMC–PDE recursion

Require: Exercise grid $\{t_n\}_{n=0}^{M}$, Euler grid containing the exercise dates, log-price grid $\{y_i\}_{i=1}^{N_S}$, volatility domain $\mathcal{D}_v$, basis ϕ, truncation level R, training size N, and evaluation size N_0

Ensure: Direct estimator $\widehat{V}_{d,0}(S_0)$ and fitted continuation surfaces $\widehat{C}_n^N$

1: Simulate N training paths of the variance factors on the Euler grid.
2: Store exercise-date volatility states and accumulate Z_n^j, I_n^j, and B_n^j over each interval $[t_n, t_{n+1}]$.
3: Set $\widehat{V}_M^N(s_i, v, w) \leftarrow h_M(s_i)$ on $\mathcal{S}_h \times \mathcal{D}_v$.
4: **for** $n = M-1, \ldots, 1$ **do**
5: **for** $j = 1, \ldots, N$ **do**
6: Compute $\{\widehat{C}_n^j(s_i)\}_{i=1}^{N_S}$ using the unshifted terminal grid g_{n+1}^j in (5), with Z_n^j included only through the multiplier $\exp(\Psi_n^j)$; do not pre-shift g_{n+1}^j by Z_n^j.
7: **end for**
8: **for** each $s_i \in \mathcal{S}_h$ **do**
9: Compute $b_n^N(s_i)$ and $a_n^N(s_i)$ by the truncated-inverse regression (7).
10: **end for**
11: Set $\widehat{C}_n^N(s_i, v, w) \leftarrow a_n^N(s_i) \cdot \phi(v, w)$ for all $s_i \in \mathcal{S}_h$.
12: Set $\widehat{V}_n^N(s_i, v, w) \leftarrow \max\{h_n(s_i), \widehat{C}_n^N(s_i, v, w)\}$ for all $s_i \in \mathcal{S}_h$.
13: **end for**
14: Form the time-zero evaluation sample: either simulate N_0 independent paths on $[t_0, t_1]$, or reuse the N training paths with $N_0 = N$ for the in-sample estimator; compute $\widehat{C}_0^{j,\mathrm{eval}}(s_i)$.
15: Set $\widehat{C}_0^{N_0}(s_i) \leftarrow N_0^{-1} \sum_{j=1}^{N_0} \widehat{C}_0^{j,\mathrm{eval}}(s_i)$.
16: Evaluate $\widehat{C}_0^{N_0}(S_0)$ from the averaged continuation grid at $y = \log S_0$, using linear interpolation if needed.
17: **return** $\widehat{V}_{d,0}(S_0) = \max\{h_0(S_0), \widehat{C}_0^{N_0}(S_0)\}$.

5 Numerical experiments

We report direct Bermudan put pricing experiments for the fixed parameter set in Table 1. The plain LSMC estimator [2] is compared with the Hybrid LSMC–PDE estimator described in Section 4, using direct prices and reference-relative errors. Standard errors and 95% confidence intervals are reported for the reference prices and used to construct the error bars in the figures.

5.1 Experimental setting and reference values

Table 1 reports the fixed gDMR parameter set and numerical settings used in the experiments.

Table 1 Parameter set and numerical settings used in the Bermudan pricing experiments.

Symbol	Value	Interpretation
S_0	100	initial asset price
v_0	0.114	initial instantaneous variance factor
v'_0	0.110	initial secondary variance factor
r	0.02	risk-free rate
T	1	maturity in years
κ_1	5.5	mean-reversion speed of v_t toward v'_t
κ_2	0.1	mean-reversion speed of v'_t toward θ
θ	0.078	long-run variance level
ξ_1	2.689	volatility scale of the first variance factor
ξ_2	0.502	volatility scale of the second variance factor
δ_1	0.94	CEV exponent of the first variance factor
δ_2	0.94	CEV exponent of the second variance factor
ρ_{12}	-0.982	correlation between the first and second Brownian drivers
ρ_{13}	-0.727	correlation between the first and third Brownian drivers
ρ_{23}	0.590	correlation between the second and third Brownian drivers
Fixed numerical setting		
N_{ex}	12	exercise dates after t_0, including maturity
K	$\{70, 80, 90, 100, 110\}$	strike set used in all experiments
N_S	301	asset-grid points in the conditional PDE solver
$(a_{\min}, a_{\max})$	$(0.30, 3.50)$	asset-range factors relative to S_0
q_v	0.999	volatility truncation quantile
$(M_{\text{ref}}, N_{\text{ref}})$	$(1200, 1{,}200{,}000)$	reference LSMC step count and path budget

The LSMC reference prices are reported in Table 2. The Hybrid LSMC–PDE grid and truncation settings are kept fixed throughout both sets of experiments.

The mean-reversion, volatility-scale, initial-variance, and correlation parameters are taken from the calibrated double mean-reverting specification of Bayer, Gatheral and Karlsmark [11]. Their calibration uses $r = 0$. In the present Bermudan pricing experiments we keep those variance and correlation parameters, set $r = 0.02$, and use $\delta_1 = \delta_2 = 0.94$ for the CEV-type generalization. Thus the numerical section is not a new calibration study; it uses a parameter set from the double mean-reverting volatility literature as a fixed test case. The correlation assumptions used in Section 3 are satisfied. In particular, $|\rho_{23}| < 1$.

The plain LSMC regressions use a cubic 16-term basis

$$1, x, y, z, x^2, y^2, z^2, xy, xz, yz, x^3, y^3, z^3, p, p^2, p^3,$$

where $x = S/S_0$, $y = v/\theta$, $z = v'/\theta$, and $p = (K - S)^+/K$. The hybrid regressions use a volatility-only cubic basis

$$1, y, z, y^2, yz, z^2, y^3, y^2z, yz^2, z^3,$$

where y and z are the normalized volatility coordinates after applying the numerical truncation to the chosen volatility rectangle D_v. The smooth compact-support assumption in Subsection 3.3 is part of the convergence statement. For the hybrid estimator, the same simulated variance-factor paths are used for fitting and for the time-zero direct average; the reported hybrid direct estimator is therefore an in-sample direct estimator in the sense described in Section 4.

For a direct estimator V and reference value V^{ref}, we report

$$\varepsilon_{\mathrm{dir}} := \frac{|V - V^{\mathrm{ref}}|}{|V^{\mathrm{ref}}|}.$$

The reference values are plain LSMC estimates computed with 1200 Euler steps and 1,200,000 direct paths. They are used as numerical reference values, not as exact prices. The reported standard errors are empirical Monte Carlo standard errors of the direct estimators, and the 95% confidence intervals are computed as the direct estimate plus or minus 1.96 standard errors. In the relative-error plots, the reference value V^{ref} is treated as fixed. The error bars are obtained by mapping the direct-estimator confidence intervals through

$$V \mapsto \frac{|V - V^{\mathrm{ref}}|}{|V^{\mathrm{ref}}|}.$$

Table 2 Direct LSMC reference values for the parameter set in Table 1.

K	Reference price	Std. error	95% confidence interval
70	2.522851	0.007103	[2.508929, 2.536773]
80	4.298933	0.009345	[4.280617, 4.317249]
90	6.942119	0.011774	[6.919042, 6.965196]
100	10.682037	0.014105	[10.654391, 10.709683]
110	15.743520	0.015757	[15.712636, 15.774404]

5.2 Fixed path budget and varying Euler steps

The first experiment keeps the direct path budget fixed and varies the number of Euler time steps. The two path budgets are 20,000 and 60,000, and both methods are evaluated at $24, 48, 72$, and 96 Euler steps. This experiment tests the sensitivity of the direct estimators to the Euler discretization.

For 20,000 paths, the LSMC reference-error range over the full step-sweep grid is 0.344%–8.645%, while the Hybrid LSMC–PDE range is 0.019%–7.459%. Figure 1 reports the strike-wise reference errors. The corresponding direct price grids are reported in Appendix B.

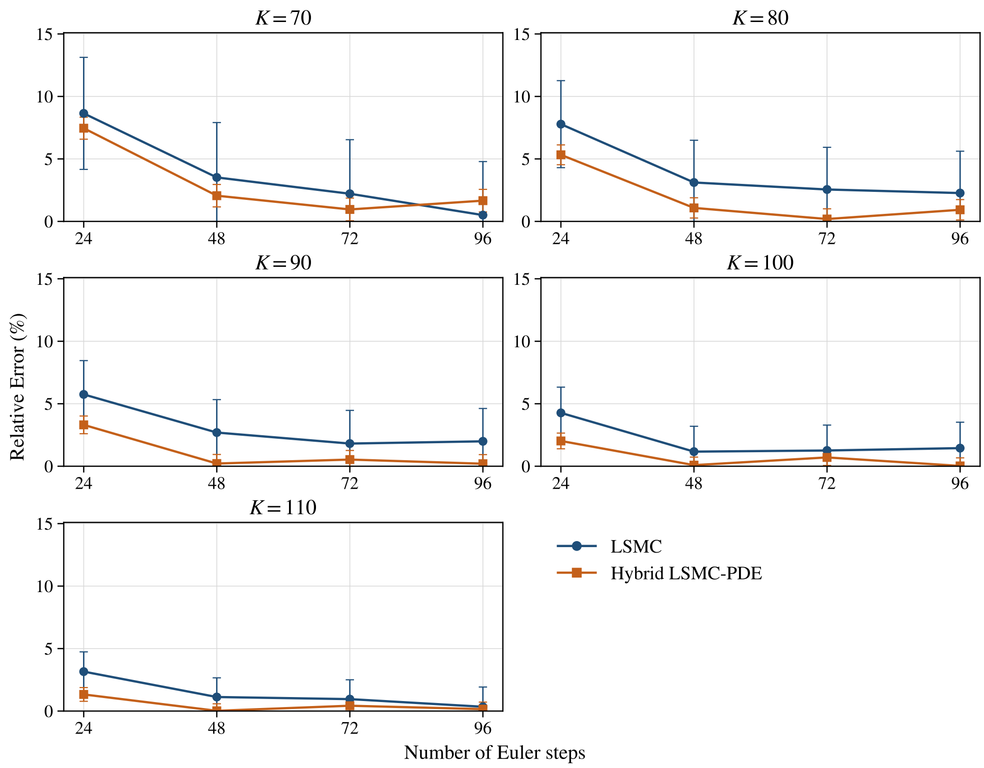

Fig. 1 Reference-relative errors for the matched 20,000-path step-sweep experiment. The reference values are the LSMC estimates in Table 2. Error bars are obtained from the 95% confidence intervals of the direct estimators.

For 60,000 paths, the same design gives an LSMC reference-error range of 0.102%–7.720% and a Hybrid LSMC–PDE range of 0.028%–7.857%. Figure 2 gives the corresponding errors, and the direct price grids are reported in Appendix B. The hybrid estimator is not uniformly better at every coarse or fine discretization point, but it gives smaller reference errors in most of the reported configurations, with the clearest differences at the intermediate step levels.

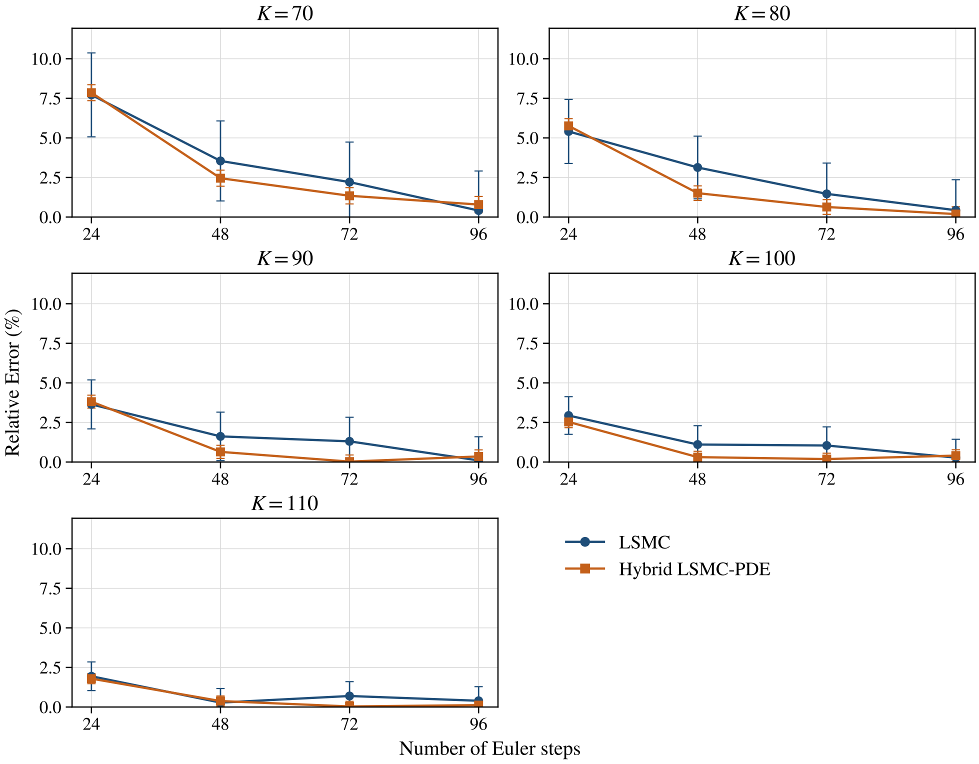

Fig. 2 Reference-relative errors for the matched 60,000-path step-sweep experiment. The reference values are the LSMC estimates in Table 2. Error bars are obtained from the 95% confidence intervals of the direct estimators.

5.3 Fixed Euler step count and varying path budget

The second experiment fixes the number of Euler time steps and varies the direct path budget. The reported path counts are $250, 1000, 5000, 10000, 20000, 40000$, and 60000. Two discretization levels, 48 and 60 Euler steps, are considered. This experiment focuses on sampling variation and regression error.

At 48 Euler steps, the at-the-money strike $K = 100$ gives an LSMC reference-error range of 1.105%–25.948% over the path grid, while the Hybrid LSMC–PDE range is 0.088%–11.118%. Figure 3 shows the path-count sweep for all five strikes. The corresponding direct prices are reported in Appendix B.

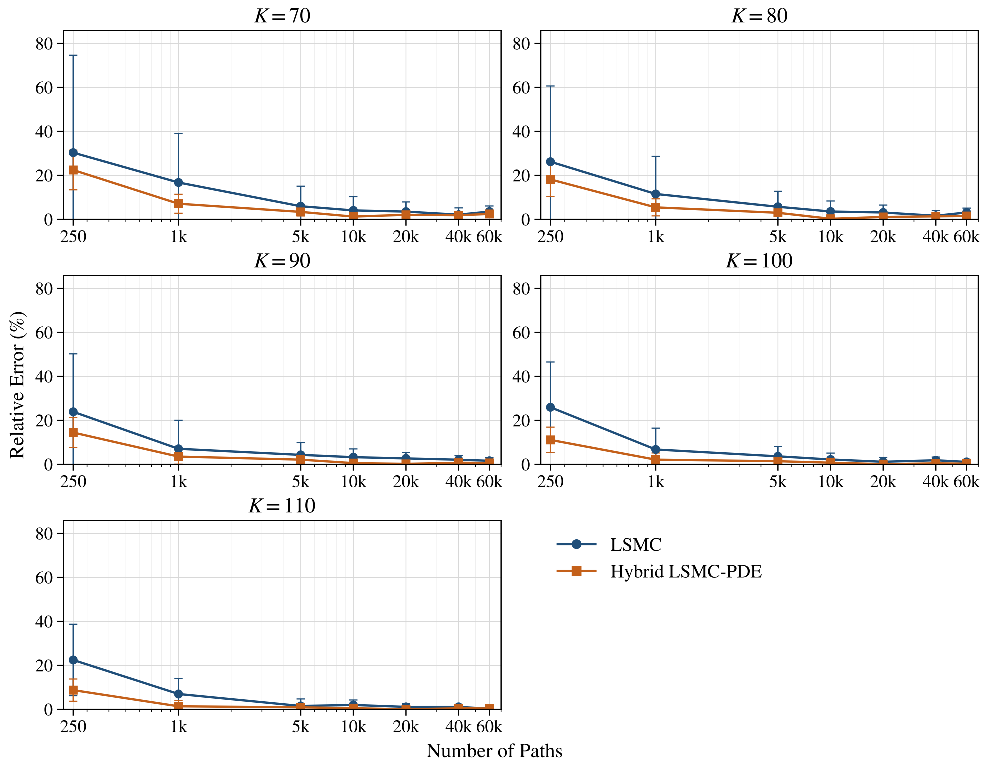

Fig. 3 Reference-relative errors for the fixed 48-step path-sweep experiment. The reference values are the LSMC estimates in Table 2. Error bars are obtained from the 95% confidence intervals of the direct estimators.

At 60 Euler steps, the corresponding at-the-money LSMC range is 1.324%–28.932%, while the Hybrid LSMC–PDE range is 0.049%–10.102%. Figure 4 shows the same path-count experiment at 60 Euler steps, with the direct prices reported in Appendix B. The reduction in reference error is most visible in the low- and moderate-path regimes, where the plain regression estimator is most affected by sampling noise.

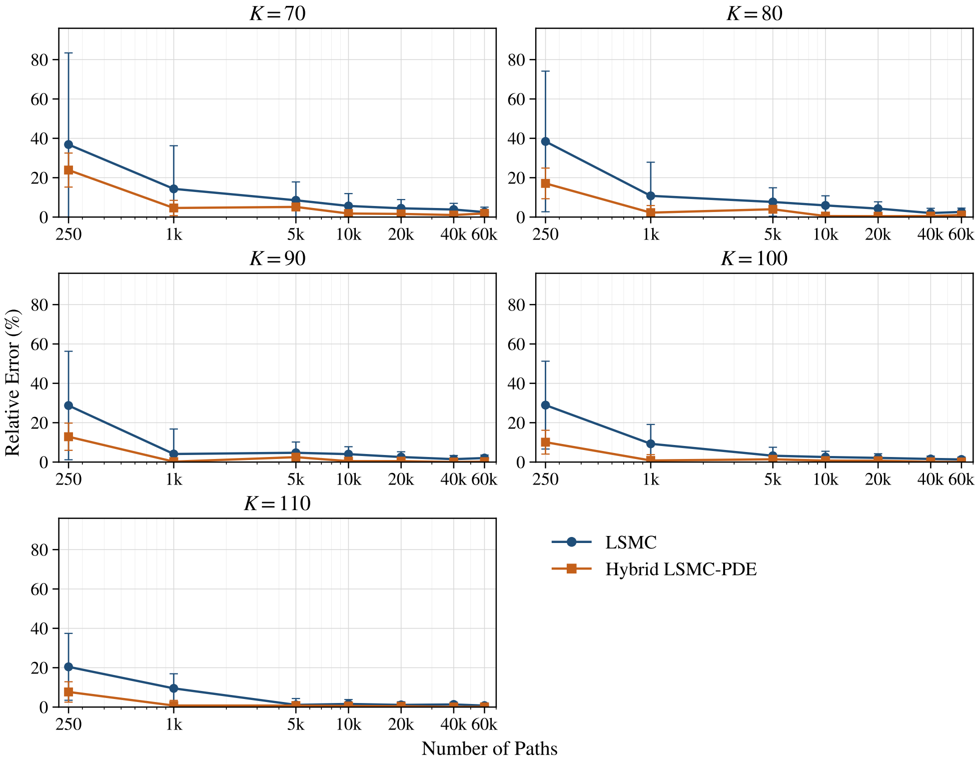

Fig. 4 Reference-relative errors for the fixed 60-step path-sweep experiment. The reference values are the LSMC estimates in Table 2. Error bars are obtained from the 95% confidence intervals of the direct estimators.

Overall, the experiments are consistent with the conditional PDE step reducing the dimension of the continuation regression. The comparisons are made against the LSMC reference values in Table 2, not exact prices. The full direct price grids are reported in Appendix B.

6 Conclusion

We studied Bermudan put pricing under a generalized Gatheral double mean-reverting stochastic-volatility model with CEV-type volatility-of-volatility coefficients. The main analytical result is that the generalized variance system is well posed: for $\delta_1, \delta_2 \in [1/2, 1]$, it has a unique strong solution and preserves the nonnegative variance state space.

The hybrid pricing construction uses the one-way coupling of the model. After conditioning on the Brownian increments driving the variance factors and projecting the asset Brownian motion onto these drivers, each conditional asset-price step reduces to a one-dimensional, possibly degenerate, problem in log-price space. The path statistics Z_n, I_n, and B_n are the model-specific quantities needed for the LSMC–PDE method and for the FFT implementation.

The numerical experiments compare the direct hybrid estimator with plain LSMC under one fixed gDMR parameter set, based on a calibrated double mean-reverting specification from the literature. The hybrid estimator gives smaller reference errors in most of the reported fixed-step and fixed-path configurations, with the clearest gains at low and moderate path budgets. The reference values are LSMC estimates computed with 1200 Euler steps and 1,200,000 paths, not exact prices. Natural extensions include low-biased estimators, broader parameter regimes, and other contract types.

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Declarations

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Conflict of interest/Competing interests

The authors declare that they have no competing interests.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Data availability

The numerical data used to generate the tables and figures are included in the manuscript source files.

Materials availability

Not applicable.

Code availability

Code is available from the corresponding author upon reasonable request.

Author contribution

Mara Kalicanin Dimitrov and Ying Ni contributed to the conceptualization, mathematical analysis, numerical design, interpretation of results, and manuscript preparation. All authors read and approved the final manuscript.

Appendix A Proof of Theorem 2.4

Proof. We work first with the auxiliary equations on $\mathbb{R}$, where the power coefficients are replaced by their positive-part extensions. Once nonnegativity has been proved, the auxiliary equations coincide with the original system on the state space $\mathcal{D}$.

Step 1: construction of v'. Define

$$b_2(x) := \kappa_2(\theta - x), \qquad \sigma_2(x) := \xi_2(x^+)^{\delta_2}, \qquad x \in \mathbb{R}.$$

The drift b_2 is globally Lipschitz and has linear growth. The diffusion coefficient σ_2 is continuous, has at most linear growth, and satisfies the Yamada–Watanabe modulus condition by Assumption 2.3. Therefore pathwise uniqueness holds for

$$\mathrm{d}v'_t = b_2(v'_t)\mathrm{d}t + \sigma_2(v'_t)\mathrm{d}W_t^{(3)}, \qquad v'_0 \geq 0;$$

see [14] and [15, Ch. 5]. Since the coefficients are continuous with linear growth, weak existence holds. Weak existence together with pathwise uniqueness yields a unique strong solution by the Yamada–Watanabe theorem.

It remains to show that the solution does not leave the nonnegative half-line. At the boundary,

$$\sigma_2(0) = 0, \qquad b_2(0) = \kappa_2\theta \geq 0.$$

Moreover, on the negative extension one has

$$\sigma_2(x) = 0, \qquad b_2(x) = \kappa_2(\theta - x) \geq 0, \qquad x \leq 0.$$

Thus the diffusion has no outward noise at the boundary and the drift points inward on the negative side. The one-dimensional stochastic invariance criterion for closed intervals, or equivalently Tanaka's formula applied to the negative part with this positive-part extension, gives

$$v'_t \geq 0, \qquad 0 \leq t \leq T, \quad \mathbb{Q}\text{-a.s.}$$

Step 2: construction of v. Having constructed v', define

$$b_1(t,x) := \kappa_1(v'_t - x), \qquad \sigma_1(x) := \xi_1(x^+)^{\delta_1}.$$

For $m \geq 1$, set

$$\tau_m := \inf\{t \geq 0 : v'_t > m\} \wedge T.$$

Since v' has continuous paths on $[0,T]$, we have $\tau_m \uparrow T$ almost surely. On $[0,\tau_m]$, the drift b_1 is progressively measurable, Lipschitz in x with constant κ_1, and satisfies

$$|b_1(t,x)| \leq \kappa_1(m + |x|).$$

The diffusion coefficient σ_1 is continuous, has at most linear growth, and satisfies the Yamada–Watanabe modulus condition. The localized one-dimensional existence and pathwise-uniqueness theorem for SDEs with progressive random coefficients, obtained by the same Yamada–Watanabe argument after localization, therefore gives a unique strong stopped solution of

$$\mathrm{d}v_t = b_1(t,v_t)\mathrm{d}t + \sigma_1(v_t)\mathrm{d}W_t^{(2)}, \qquad t \leq \tau_m.$$

The possible correlation between $W^{(2)}$ and $W^{(3)}$ is immaterial here: pathwise uniqueness compares solutions driven by the same Brownian vector and the same realized input path v'. By pathwise uniqueness, the stopped solutions are compatible on overlaps. Letting $m \to \infty$ gives a unique strong solution for v on $[0,T]$.

The same invariance argument gives nonnegativity. Since $v'_t \geq 0$,

$$b_1(t,0) = \kappa_1 v'_t \geq 0, \qquad \sigma_1(0) = 0.$$

On the negative extension,

$$b_1(t,x) = \kappa_1(v'_t - x) \geq 0, \qquad \sigma_1(x) = 0, \qquad x \leq 0.$$

Hence the closed half-line is invariant and

$$v_t \geq 0, \qquad 0 \leq t \leq T, \quad \mathbb{Q}\text{-a.s.}$$

Step 3: construction of S. The process v is continuous and finite on the compact interval $[0,T]$. Therefore

$$\int_0^T v_s \,\mathrm{d}s < \infty \qquad \mathbb{Q}\text{-a.s.}$$

The asset equation is linear in S, and its unique strong solution is

$$S_t = S_0 \exp\biggl(\int_0^t \Bigl(r - \tfrac{1}{2}v_s\Bigr)\mathrm{d}s + \int_0^t \sqrt{v_s}\,\mathrm{d}W_s^{(1)}\biggr).$$

Thus $S_t > 0$ for all $t \in [0,T]$ whenever $S_0 > 0$.

The auxiliary positive-part extension is inactive along the constructed variance paths, because $v_t, v_t' \geq 0$. Hence the constructed solution solves the original nonnegative-state system (1). Pathwise uniqueness for the auxiliary triangular system gives pathwise uniqueness for (1).

Finally, existence and pathwise uniqueness for every deterministic initial state imply uniqueness in law. Since the coefficients are time homogeneous, the standard strong Markov property for well-posed time-homogeneous SDEs gives that the solution is a time-homogeneous strong Markov process on $\mathcal{D}$. □

Appendix B Direct prices for the numerical experiments

This appendix reports the direct Bermudan put prices corresponding to the numerical experiments in Section 5. All entries are rounded to three decimal places. The label LSMC denotes the plain least-squares Monte Carlo estimator, and Hybrid denotes the Hybrid LSMC–PDE estimator.

B.1 Reference prices

Table B1 Direct reference prices for the parameter set in Table 1. Prices are rounded to three decimal places.

Reference setting	$K = 70$	$K = 80$	$K = 90$	$K = 100$	$K = 110$
1200 steps, 1,200,000 paths	2.523	4.299	6.942	10.682	15.744

Table B2 Direct prices in the fixed-path step-sweep experiment with 20,000 paths. Prices are rounded to three decimal places.

Method/step	24	48	72	96
LSMC $K = 70$	2.741	2.612	2.579	2.536
Hybrid $K = 70$	2.711	2.575	2.547	2.565
LSMC $K = 80$	4.634	4.433	4.409	4.396
Hybrid $K = 80$	4.528	4.345	4.307	4.339
LSMC $K = 90$	7.341	7.129	7.068	7.080
Hybrid $K = 90$	7.172	6.957	6.905	6.956
LSMC $K = 100$	11.138	10.807	10.816	10.837
Hybrid $K = 100$	10.898	10.673	10.607	10.679
LSMC $K = 110$	16.241	15.920	15.894	15.798
Hybrid $K = 110$	15.953	15.747	15.676	15.768

Table B3 Direct prices in the fixed-path step-sweep experiment with 60,000 paths. Prices are rounded to three decimal places.

Method/step	24	48	72	96
LSMC $K = 70$	2.718	2.612	2.579	2.533
Hybrid $K = 70$	2.721	2.585	2.557	2.543
LSMC $K = 80$	4.531	4.434	4.362	4.318
Hybrid $K = 80$	4.547	4.364	4.326	4.307
LSMC $K = 90$	7.195	7.055	7.033	6.949
Hybrid $K = 90$	7.207	6.987	6.940	6.918
LSMC $K = 100$	10.996	10.800	10.794	10.712
Hybrid $K = 100$	10.953	10.715	10.662	10.639
LSMC $K = 110$	16.050	15.788	15.853	15.806
Hybrid $K = 110$	16.026	15.803	15.749	15.725

Table B4 Direct prices in the fixed 48-step path-sweep experiment. Prices are rounded to three decimal places.

Method/path count	250	1,000	5,000	10,000	20,000	40,000	60,000
LSMC $K = 70$	3.288	2.946	2.673	2.625	2.612	2.576	2.612
Hybrid $K = 70$	3.089	2.702	2.608	2.554	2.575	2.570	2.585
LSMC $K = 80$	5.425	4.795	4.545	4.452	4.433	4.369	4.434
Hybrid $K = 80$	5.079	4.533	4.427	4.310	4.345	4.357	4.364
LSMC $K = 90$	8.601	7.433	7.241	7.169	7.129	7.088	7.055
Hybrid $K = 90$	7.946	7.187	7.090	6.903	6.957	6.992	6.987
LSMC $K = 100$	13.454	11.404	11.072	10.918	10.807	10.881	10.800
Hybrid $K = 100$	11.870	10.910	10.833	10.601	10.673	10.730	10.715
LSMC $K = 110$	19.278	16.843	15.985	16.057	15.920	15.921	15.788
Hybrid $K = 110$	17.120	15.962	15.889	15.660	15.747	15.819	15.803

Table B5 Direct prices in the fixed 60-step path-sweep experiment. Prices are rounded to three decimal places.

Method/path count	250	1,000	5,000	10,000	20,000	40,000	60,000
LSMC $K = 70$	3.452	2.884	2.738	2.665	2.635	2.619	2.587
Hybrid $K = 70$	3.125	2.639	2.653	2.568	2.563	2.549	2.570
LSMC $K = 80$	5.950	4.762	4.629	4.553	4.484	4.387	4.411
Hybrid $K = 80$	5.034	4.396	4.470	4.321	4.318	4.321	4.342
LSMC $K = 90$	8.936	7.227	7.270	7.222	7.120	7.045	7.084
Hybrid $K = 90$	7.836	6.957	7.115	6.910	6.910	6.939	6.957
LSMC $K = 100$	13.773	11.673	11.026	10.955	10.910	10.850	10.823
Hybrid $K = 100$	11.761	10.596	10.831	10.604	10.610	10.666	10.677
LSMC $K = 110$	18.962	17.240	15.920	15.991	15.918	15.952	15.860
Hybrid $K = 110$	16.953	15.621	15.863	15.659	15.680	15.755	15.757

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